IAT New Test 5

Question paper, answer key
and worked solutions

MANUAL REVIEW DRAFT
2027 preparation using the verified 2026 pattern and stored 2026 syllabus.
The 2027 brochure has not been verified. Not approved for student publication.

60	180	240
single-correct questions	minutes	maximum marks

Subject	Question numbers	Marks
Biology	1-15	60
Chemistry	16-30	60
Mathematics	31-45	60
Physics	46-60	60

Instructions

1. There are 60 single-correct questions: Biology 1–15, Chemistry 16–30, Mathematics 31–45 and Physics 46–60.
2. Each question has exactly one correct option. Correct: +4; incorrect: -1; unanswered: 0. All questions are available to attempt; there is no best-three-subject scoring rule.
3. Use 180 minutes. An on-screen non-programmable scientific calculator is part of the 2026 examination conditions; personal calculators are not allowed under that scheme.
4. Treat stated data and idealisations as exact unless an approximation is explicitly supplied. Log denotes base 10 and ln denotes the natural logarithm.
5. The review PDF contains the question paper first, followed by a separate answer key and worked solutions. Keep the solutions hidden during a timed attempt.

Fixed authored order. Figures and IDs must remain linked during review.
Revised 2026-10-31 · IAT27_TEST_05 · Original AI-assisted content.

QUESTION PAPER / B

Biology

Questions 1-15 · 15 questions · 60 marks

Q1

+4 / -1

A seed develops from an ovule enclosed in an ovary. Mature seed has one cotyledon and parallel leaf venation. Which assignment combines the reproductive and vegetative evidence?

A) Monocot angiosperm

B) Dicot gymnosperm

C) Monocot gymnosperm

D) Dicot angiosperm

Q2

+4 / -1

A frog's heart has two atria and one ventricle. If flow from the pulmonary veins to the left atrium is blocked, which immediate oxygenated-blood entry is most directly reduced?

A) Into the right atrium from systemic veins

B) Into the left atrium from lungs and skin

C) Into the ventricle from the right atrium only

D) Into the truncus arteriosus from the ventricle

Q3

+4 / -1

In cell fractionation, fraction X is rich in catalase but poor in chlorophyll and mitochondrial cytochrome oxidase. Which organelle is enriched?

A) Chloroplast

B) Mitochondrion

C) Peroxisome

D) Nucleus

Q4

+4 / -1

At constant light, sudden stomatal closure in a well-watered C_3 leaf most directly lowers which entry flux?

A) Mineral ions into roots

B) Sucrose into phloem

C) Oxygen into mitochondria only

D) CO_2 from air into intercellular spaces

Q5

+4 / -1

At a fixed tissue pO_2, curve B in the figure has lower haemoglobin saturation than curve A. If B is measured after an increase in CO_2 and temperature, what does the shift favour?

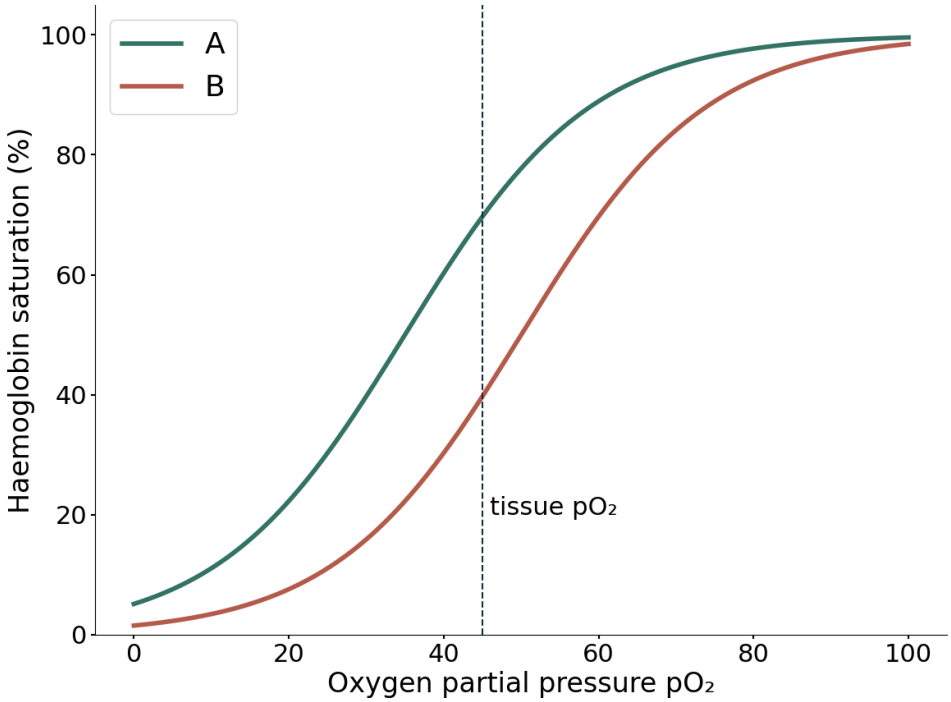


Figure for Q5

A) Greater oxygen unloading to tissue

B) Greater oxygen affinity at all pO_2

C) Complete loss of haemoglobin

D) Oxygen release only in the lungs

Q6

+4 / -1

One primary spermatocyte completes meiosis normally. How many haploid spermatids are produced, and what is each one's chromosome count in humans?

A) Two, 23

B) Four, 23

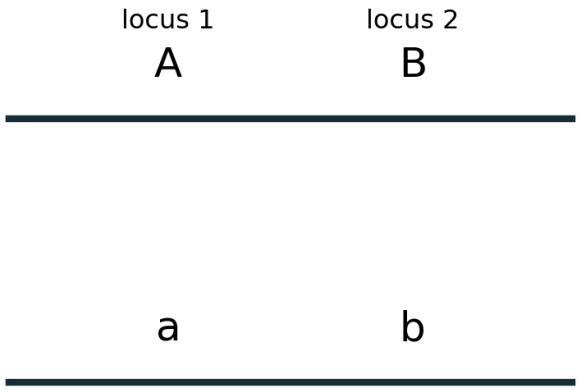
C) Four, 46

D) Eight, 23

Q7

+4 / -1

The figure shows homologues AB/ab in a double heterozygote. Which two gametes are recombinant after a crossover between loci A and B?



Homologous chromosomes before crossing over

Figure for Q7

A) The unchanged parental pair AB and ab

B) AB plus Ab

C) Ab and aB

D) The pair ab and aB with parental ab

Q8

+4 / -1

A microbial preparation selectively dissolves fibrin clots after myocardial infarction. Which product and source pairing is appropriate?

A) Cyclosporin A from Trichoderma polysporum

B) Statins from Monascus purpureus

C) Penicillin from Penicillium notatum

D) Streptokinase from Streptococcus

Q9

+4 / -1

A crop expresses a Bt toxin gene active against a particular insect larva. What determines whether the larva is susceptible after eating the crop?

A) Activation and binding of toxin in the larval gut

B) Whether the crop is a monocot

C) Presence of chlorophyll in the insect

D) The insect's ability to photosynthesise

Q10

+4 / -1

Two forests have equal species richness, but forest X contains many endemic species threatened by habitat loss and forest Y mostly widespread species. Which observation strengthens the case for prioritising X for in-situ conservation?

A) Every species in Y is also endemic

B) Loss of X would remove unique lineages from their native habitat

C) Equal richness guarantees equal extinction risk

D) Ex-situ preservation always maintains the entire ecosystem

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Q11

+4 / -1

A father has an X-linked recessive trait and the mother is homozygous unaffected. Which statement about their daughters is correct?

A) All daughters are affected

B) Half the daughters are affected

C) All daughters inherit the father's variant X and are carriers

D) No daughter inherits any paternal X

Q12

+4 / -1

A freely filtered solute has plasma concentration 2 mg/mL, urine concentration 40 mg/mL and urine flow 1 mL/min. What is its clearance?

A) 5 mL/min

B) 10 mL/min

C) 80 mL/min

D) 20 mL/min

Q13

+4 / -1

A neurotransmitter opens ligand-gated Na^+ channels on the postsynaptic membrane. What immediate electrical effect is most likely if Na^+ enters down its electrochemical gradient?

A) Depolarisation of the postsynaptic membrane

B) Hyperpolarisation from K^+ efflux

C) No membrane-potential change because ligand binding consumes ATP

D) Repolarisation from chloride exit

Q14

+4 / -1

In the figure, enzyme curve I is shifted to higher substrate concentration at half-maximal speed but reaches the same maximum rate as control. Curve II reaches a lower maximum even at high substrate. Which assignment fits?

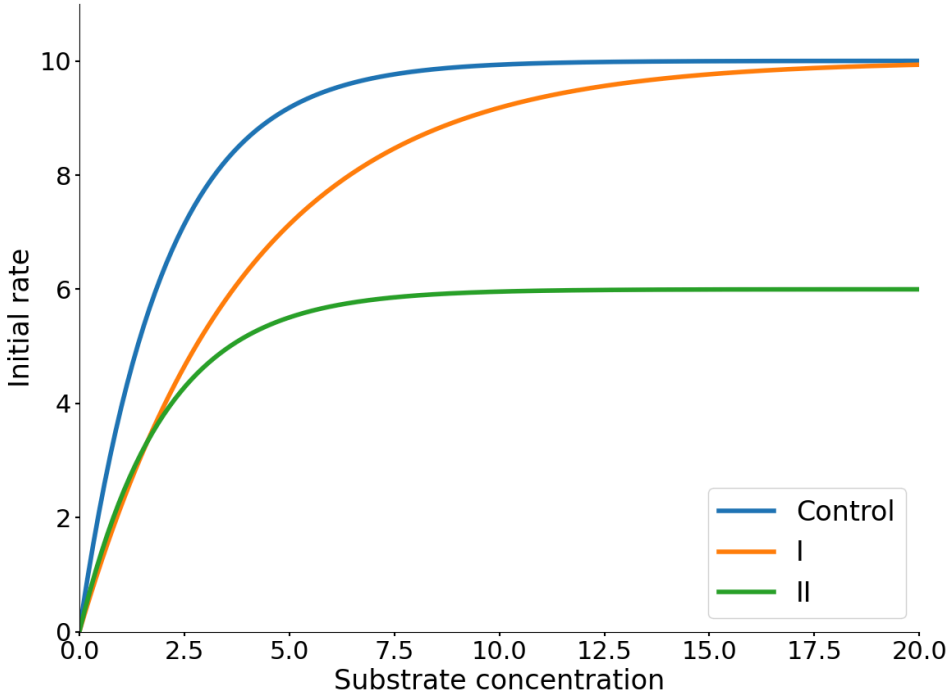


Figure for Q14

A) I and II both competitive

B) I competitive; II noncompetitive

C) I noncompetitive; II competitive

D) Neither shows inhibition

Q15

+4 / -1

After ovulation, corpus luteum progesterone keeps the uterine lining secretory. If no implantation occurs, what event most directly precedes menstruation?

A) Continued rise of progesterone

B) Fertilisation in the uterus

C) Degeneration of corpus luteum and fall in progesterone

D) Permanent arrest of endometrial shedding

Chemistry

Questions 16-30 · 15 questions · 60 marks

Q16

+4 / -1

A neutral molecule XF_4 has six electron domains around its central atom X, of which four are X–F bonds. Assuming equivalent lone pairs and minimal repulsion, what molecular shape and central lone-pair count follow?

A) Square planar, two lone pairs

B) Tetrahedral, two lone pairs

C) Seesaw, one lone pair

D) Square pyramidal, one lone pair

Q17

+4 / -1

For the same metal ion, replacing six monodentate ligands by three bidentate ligands can make the complex more stable. Which explanation is most defensible when donor-atom chemistry is comparable?

A) The metal oxidation state must decrease

B) The chelate effect often gives a favourable entropy change from ligand exchange

C) Bidentate ligands have no coordination bonds

D) Stability is controlled solely by the number of metal ions

Q18

+4 / -1

The figure shows a broadly decreasing ionic radius across trivalent lanthanoid ions as atomic number grows. Which cause best fits the trend?

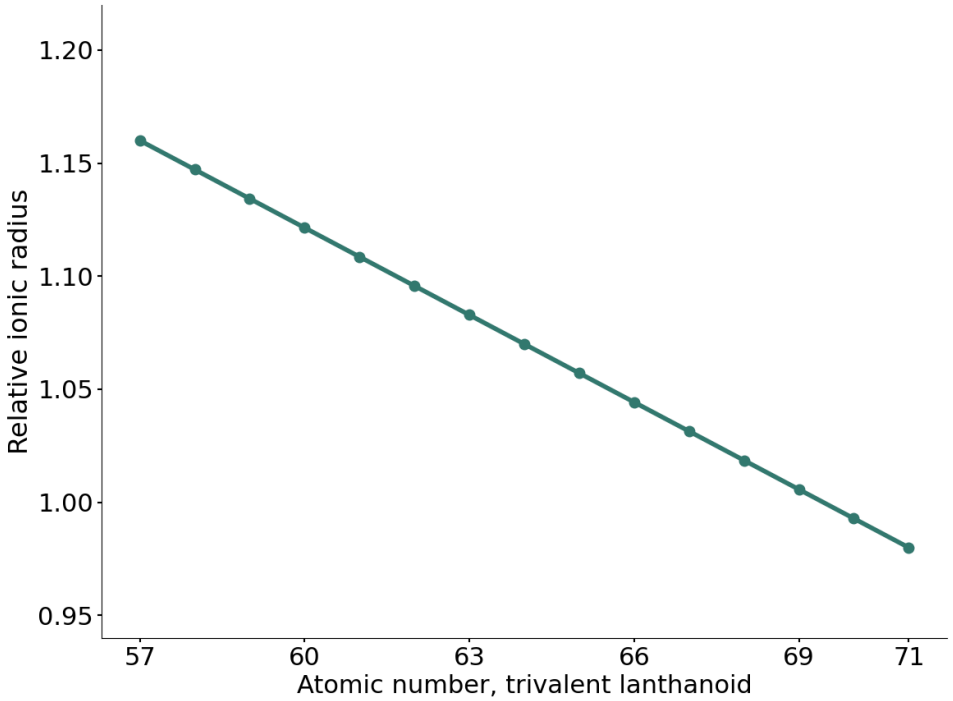


Figure for Q18

A) Every ion loses an additional proton

B) 4f electrons shield nuclear charge exceptionally well

C) All ions become divalent

D) Added 4f electrons shield poorly, increasing effective nuclear attraction

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Q19

+4 / -1

In hydrogen, compare wavelengths of the 3→2 and 2→1 emission lines. What is λ(3→2)/λ(2→1)?

A) 5/27

B) 9/4

C) 27/5

D) 3/2

Q20

+4 / -1

Among O^{2-}, F^{-}, Na^{+} and Mg^{2+}, all with ten electrons, which order of decreasing ionic radius is expected?

A) $O^{2-} > F^{-} > Na^{+} > Mg^{2+}$

B) $Mg^{2+} > Na^{+} > F^{-} > O^{2-}$

C) $F^{-} > O^{2-} > Na^{+} > Mg^{2+}$

D) $Na^{+} > Mg^{2+} > O^{2-} > F^{-}$

Q21

+4 / -1

100 mL of 0.40 M Na_2CO_3 is diluted to 400 mL. A 100 mL aliquot is reacted with 15 mmol HCl. Assuming complete reaction $Na_2CO_3 + 2HCl \rightarrow 2NaCl + H_2O + CO_2$, what mass of CO_2 is evolved?

A) 0.44 g

B) 0.33 g

C) 0.66 g

D) 1.76 g

Q22

+4 / -1

Phenol is more acidic than cyclohexanol under comparable conditions. Which comparison of their conjugate bases best explains this?

A) Cyclohexoxide is resonance stabilised over an aromatic ring

B) Phenoxide cannot carry negative charge on oxygen

C) Phenoxide delocalises negative charge into the ring whereas cyclohexoxide lacks that resonance

D) Phenol releases a hydride ion instead of a proton

Q23

+4 / -1

In $CH_3COOH + C_2H_5OH \rightleftharpoons CH_3COOC_2H_5 + H_2O$, what change favours more ester at equilibrium without changing temperature?

A) Add water

B) Remove ethanol

C) Add ester

D) Continuously remove water

Q24

+4 / -1

Anisole (C_6H_5–O–CH_3) is heated with excess HI. Which main organic products arise from cleavage of its ether bond?

A) Phenol and CH_3I

B) Iodobenzene and methanol

C) Benzene and formaldehyde

D) Cyclohexanol and CH_3I

Q25

+4 / -1

Direct treatment of aniline with bromine water can give 2,4,6-tribromoaniline. Which sequence best favours mainly the para monobromo derivative after deprotection?

A) Oxidise the ring, then reduce it

B) Acetylate –NH_2, brominate, then hydrolyse the amide

C) Brominate three times, then add more bromine

D) Remove the amino group before electrophilic substitution

Q26

+4 / -1

In double-stranded DNA, 30% of all bases are adenine. What fraction are guanine?

A) 10%

B) 30%

C) 20%

D) 40%

Q27

+4 / -1

For a liquid with xA=xB=0.5, pA°=100 kPa and pB°=50 kPa, total vapour pressure is measured as 60 kPa. Relative to ideal Raoult behaviour, what deviation and interaction pattern does this support?

A) +15 kPa; weaker unlike attractions

B) −15 kPa; weaker unlike attractions

C) +15 kPa; stronger unlike attractions

D) −15 kPa; stronger unlike attractions

Q28

+4 / -1

The figure gives an aqueous $CuSO_4$/NaCl cell with inert electrodes and qualitative discharge-preference bars for the competing species at each electrode under the stated conditions. Which products form first?

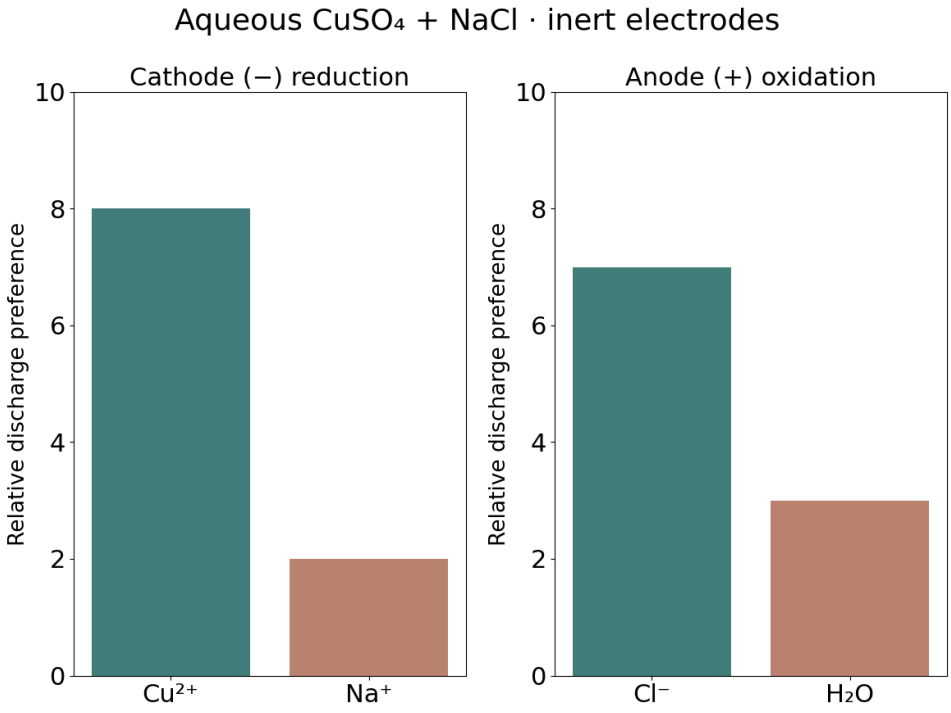


Figure for Q28

A) Cu(s) at cathode and Cl_2(g) at anode

B) Na(s) at cathode and O_2(g) at anode

C) Cu(s) at cathode and H_2(g) at anode

D) H_2(g) at cathode and SO_2(g) at anode

Q29

+4 / -1

Two Arrhenius reactions have the same pre-exponential factor at one fixed temperature T. Reaction X has activation energy E, while Y has E+RT ln 2. What is kY/kX?

A) 2

B) 1/2

C) 1

D) 1/4

Q30

+4 / -1

For A(g)+B(g) ⇌ C(g), Kc=2 M^{-1}. At an instant [A]=[B]=1 M and [C]=4 M. In which direction will the reaction initially shift?

A) Forward because reactants are equal

B) No shift because total concentration is 6 M

C) Forward because Qc exceeds Kc

D) Reverse because Qc=4 M^{-1} exceeds Kc

QUESTION PAPER / M

Mathematics

Questions 31-45 · 15 questions · 60 marks

Q31

+4 / -1

For what value of c does $x^2+y^2-4x+6y+c=0$ touch the x-axis at exactly one point?

A) 4

B) 9

C) 13

D) -4

Q32

+4 / -1

On which open intervals is $f(x)=x^3-3x$ strictly increasing?

A) $(-1,1)$ only

B) $(-\infty,-1)$ and $(1,\infty)$

C) $(-\infty,1)$ only

D) All real numbers except ±1

Q33

+4 / -1

For $a>0$, $\int_0^a 2x/(1+x^2)^2\,dx=3/4$. What is a?

A) 1

B) 2

C) $\sqrt{3}$

D) $\sqrt{2}$

Q34

+4 / -1

For $f(t)=\sqrt{(t-1)}$ and $g(x)=1/(3-x)$, what is the real domain of $f(g(x))$?

A) $x<3$

B) $x\geq2$

C) $x\leq2$

D) $[2,3)$

Q35

+4 / -1

A 3×3 matrix A has determinant 5. Matrix B is formed by replacing row 2 by row 2+3(row 1). What is det B?

A) 5

B) 15

C) 20

D) 0

Q36

+4 / -1

Box A is chosen with probability 2/3 and holds 2 red, 1 blue balls. Box B is chosen with probability 1/3 and holds 3 red, 1 blue balls. Two balls are drawn without replacement from the chosen box and both are red. What is the posterior probability of box B?

A) 1/3

B) 3/7

C) 1/2

D) 4/7

Q37

+4 / -1

How many distinct arrangements of the letters of MISSISSIPPI place all four I's consecutively?

A) 1680

B) 420

C) 3360

D) 840

Q38

+4 / -1

If C(n,3)=C(n,4) for integer n≥4, what is n?

A) 6

B) 8

C) 7

D) 9

Q39

+4 / -1

If the sum of the first n terms is $S_n=3n^2+2n$ for $n\geq1$ and $S_0=0$, what is the fifth term?

A) 29

B) 25

C) 35

D) 31

Q40

+4 / -1

A real a makes L=lim as x→0 of [e^(2x)−1−2x−a x²]/[e^x−1−x−x²/2] finite. What is the pair (a,L)?

A) (4,4)

B) (2,4)

C) (4,8)

D) (2,8)

Q41

+4 / -1

The family of curves $y^2=4a(x+b)$, with nonzero a and arbitrary b, has which differential equation after eliminating both constants?

A) $yy''-(y')^2=0$

B) $yy''+(y')^2=0$

C) $y''=0$

D) $yy'=0$

Q42

+4 / -1

A line has direction vector (1,2,2). What is the sine of its acute angle with the xy-plane?

A) 1/3

B) √5/3

C) 2/3

D) 1

Q43

+4 / -1

Points A(1,2,3), B(a,4,5) and C(5,6,7) are collinear with B between A and C. What are (a, AB:BC)?

A) (2, 1:1)

B) (3, 1:2)

C) (4, 1:1)

D) (3, 1:1)

Q44

+4 / -1

Values 0,2,4 occur with frequencies 1,2,1. What is mean deviation about the mean?

A) 1

B) 2

C) 1/2

D) 4/3

Q45

+4 / -1

How many distinct x in [0,4π] satisfy sin x=cos(2x)?

A) 4

B) 5

C) 6

D) 8

Physics

Questions 46-60 · 15 questions · 60 marks

Q46

+4 / -1

A particle starts at rest, accelerates at +2 m/s² for 2 s, then at −1 m/s² for 4 s. What displacement occurs in the full 6 s?

A) 12 m

B) 8 m

C) 4 m

D) 16 m

Q47

+4 / -1

A 60 kg person on a scale in an elevator reads 720 N while accelerating upward. With g=10 m/s², what will the scale read if acceleration reverses with the same magnitude?

A) 600 N

B) 480 N

C) 720 N

D) 360 N

Q48

+4 / -1

As shown, a block starts from rest 5 m above a frictionless track, then traverses 10 m of horizontal rough track with μk=0.20. For g=10 m/s², what is v² at the rough segment's end?

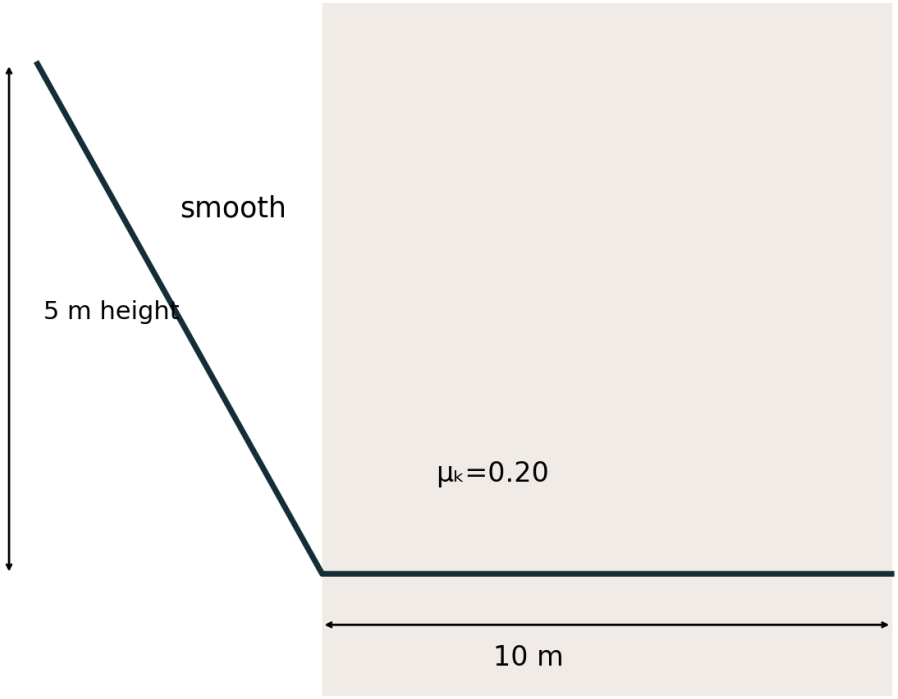


Figure for Q48

A) 100 m²/s²

B) 80 m²/s²

C) 60 m²/s²

D) 20 m²/s²

Q49

+4 / -1

A rigid rotor's applied torque and moment of inertia are each doubled. What happens to its angular acceleration, assuming both torques act in the same rotational sense?

A) It quadruples

B) It doubles

C) It halves

D) It remains unchanged

Q50

+4 / -1

For a uniform-density spherical planet of radius R, what is g at r=R/2 relative to surface value g(R)?

A) g(R)/2

B) g(R)/4

C) 2g(R)

D) 4g(R)

Q51

+4 / -1

Two spheres of the same material settle slowly in one viscous liquid under Stokes drag. If one radius is twice the other, what is the ratio of their terminal speeds (larger/smaller)?

A) 2

B) 4

C) 8

D) 16

Q52

+4 / -1

An ideal gas with γ=5/3 expands reversibly and adiabatically to twice its initial volume. What is Tf/Ti?

A) 1/2

B) 2^(2/3)

C) 2^(−2/3)

D) 2^(−5/3)

Q53

+4 / -1

A closed Gaussian surface encloses +q, −q and +2q. What is its net outward electric flux?

A) Zero

B) q/ε_0

C) $4q/\varepsilon_0$

D) $2q/\varepsilon_0$

Q54

+4 / -1

Capacitors C and 2C are connected in series to an ideal battery of voltage V. What is the magnitude of charge on the 2C capacitor?

A) 2CV/3

B) CV/3

C) CV

D) 2CV

Q55

+4 / -1

The figure shows a magnetic dipole p rotating slowly from aligned (0°) to 60° relative to uniform B. What work must an external agent do quasistatically?

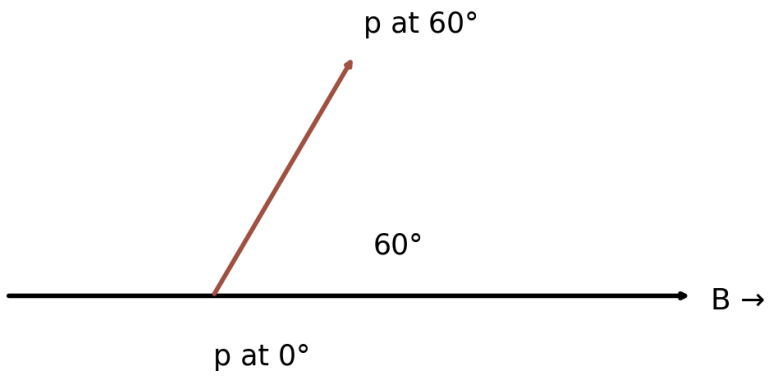


Figure for Q55

A) −pB/2

B) +pB/2

C) +pB

D) Zero

Q56

+4 / -1

For two coils with mutual inductance 0.50 H, primary current grows at +2 A/s in the chosen positive direction. What is induced secondary emf under the matching positive flux convention?

A) +1 V

B) −2 V

C) −1 V

D) +2 V

Q57

+4 / -1

A thin convex lens of focal length 10 cm has an object 30 cm to its left. A plane mirror is placed d cm to its right, before the first-pass image at 15 cm. For the reflected rays emerging back through the lens to be parallel, what is d?

A) 5 cm

B) 10 cm

C) 15 cm

D) 12.5 cm

Q58

+4 / -1

The figure shows two-beam interference with Imax=9 units and Imin=1 unit. What is the fringe visibility?

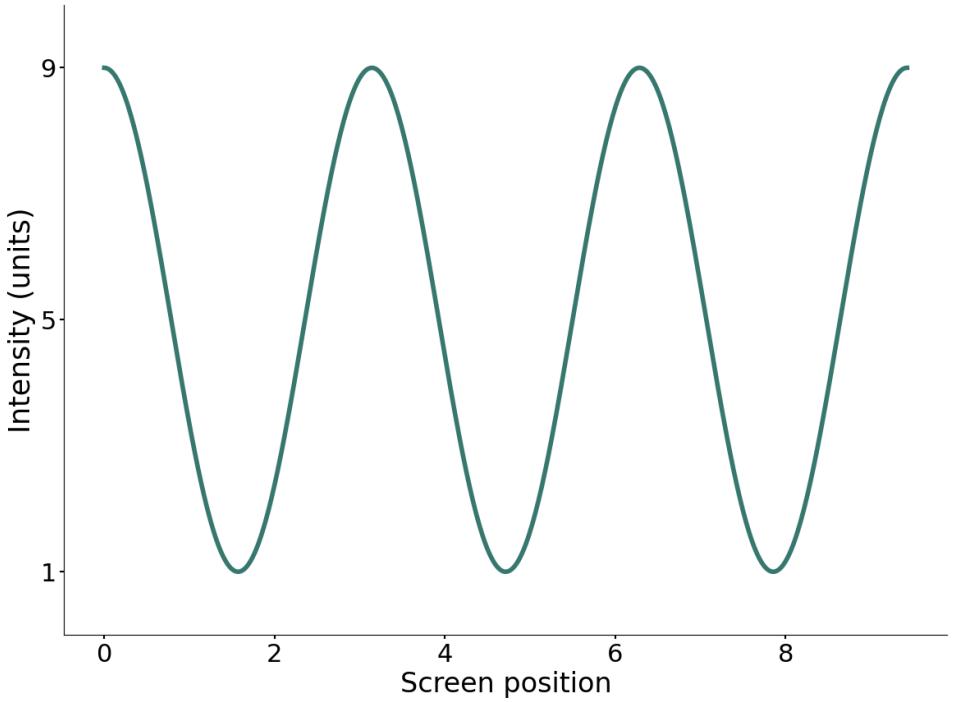


Figure for Q58

A) 1/2

B) 1

C) 1/5

D) 4/5

Q59

+4 / -1

For a hydrogen emission line terminating at n=2, its wavenumber is 3R/16 (R is Rydberg constant). From which initial level did the electron fall?

A) n=3

B) n=4

C) n=5

D) n=6

Q60

+4 / -1

Two pure radioisotopes initially have equal activities. Their half-lives are 2 h and 4 h, respectively. What is the ratio of first to second activities after 4 h?

A) 1:4

B) 2:1

C) 1:2

D) 1:1

Answer key

One correct option per question. Question and option order are fixed.

Biology

Q	Answer
1	**A**
2	**B**
3	**C**
4	**D**
5	**A**
6	**B**
7	**C**
8	**D**
9	**A**
10	**B**
11	**C**
12	**D**
13	**A**
14	**B**
15	**C**

Chemistry

Q	Answer
16	**A**
17	**B**
18	**D**
19	**C**
20	**A**
21	**B**
22	**C**
23	**D**
24	**A**
25	**B**
26	**C**
27	**D**
28	**A**
29	**B**
30	**D**

Mathematics

Q	Answer
31	**A**
32	**B**
33	**C**
34	**D**
35	**A**
36	**B**
37	**D**
38	**C**
39	**A**
40	**D**
41	**B**
42	**C**
43	**D**
44	**A**
45	**C**

Physics

Q	Answer
46	**A**
47	**B**
48	**C**
49	**D**
50	**A**
51	**B**
52	**C**
53	**D**
54	**A**
55	**B**
56	**C**
57	**D**
58	**D**
59	**B**
60	**C**

Reviewer checklist

Accuracy and single correct answer · Syllabus fit · Distractor quality
Diagram labels · Solution clarity · Difficulty and timed attempt

Reviewer: _____ Date: _____

Decision / notes: _____

Editorial difficulty mix: 9 easy / 42 medium / 9 hard. These labels require human review and are not official item statistics.

Biology

Method, reasoning and common errors

Q1

Biology · medium

Diversity In the Living World / Angiosperm seed character

Answer: (A) Monocot angiosperm.

Why: An enclosed ovule identifies angiosperms; one cotyledon and parallel venation independently support the monocot group. Both sets of evidence must agree. The enclosed ovule identifies the flowering-plant lineage before any vegetative clue is used. The single embryonic cotyledon and parallel venation independently point to the monocot side of that lineage.

Why not the others: (B) and (C) contradict ovary enclosure; (D) contradicts cotyledon number and typical venation.

Key point: Use floral origin and embryo characters together.

Q2

Biology · hard

Structural Organisation In Plants And Animals / Frog heart and circulation

Answer: (B) Into the left atrium from lungs and skin.

Why: Oxygenated blood returns from respiratory surfaces by pulmonary veins to the left atrium; blocking that return reduces this inflow first. Downstream ventricular output may change later, but it is not the directly blocked pathway. The left atrium is the receiving chamber for oxygenated return from respiratory surfaces. Blocking that vessel does not directly obstruct deoxygenated systemic venous inflow into the right atrium, although later cardiac output may change.

Why not the others: (A) is systemic venous return; (C) is a distinct atrial route; (D) is ventricular outflow.

Key point: Trace the vessel to its receiving chamber before predicting effects.

Q3

Biology · medium

Cell: Structure And Functions / Organelle fractionation marker enzymes

Answer: (C) Peroxisome.

Why: Catalase decomposes hydrogen peroxide generated by peroxisomal oxidases. Low chlorophyll and cytochrome oxidase help exclude chloroplast and mitochondrial contamination as the principal source. Catalase detoxifies peroxide inside peroxisomes, making it a functional marker rather than a generic protein. Low chlorophyll and respiratory oxidase are independent negative controls against chloroplast or mitochondrial enrichment.

Why not the others: (A) is marked by chlorophyll; (B) by respiratory-chain oxidases; (D) contains chromosomes rather than this characteristic catalase activity.

Key point: Identify an organelle by a marker plus negative controls.

Q4

Biology · easy

Plant Physiology / Stomatal closure and gas exchange

Answer: (D) CO_2 from air into intercellular spaces.

Why: Stomata are the principal gas-exchange pores; closing them raises diffusion resistance for external CO_2, limiting substrate supply to photosynthetic cells. Other processes can respond secondarily. The barrier is at the leaf surface where gas exchange with air occurs. Reduced CO_2 supply can subsequently reduce carboxylation, but the question asks for the *immediate entry flux*, not a downstream metabolic rate.

Why not the others: (A) occurs in roots; (B) is phloem loading; (C) is not the uniquely direct stomatal entry flux.

Key point: Locate the anatomical barrier before inferring metabolic consequences.

Q5

Biology · medium

Human Physiology / Haemoglobin oxygen dissociation shift

Answer: (A) Greater oxygen unloading to tissue.

Why: Higher CO_2 and temperature shift the dissociation relation rightward, lowering saturation at the same tissue oxygen pressure. More bound oxygen can therefore be released during passage through active tissue. Compare the two curves at the dashed common tissue pO_2, not at different x positions. Curve B leaves less oxygen bound after tissue passage; the saturation difference is oxygen available for unloading.

Why not the others: (B) reverses the affinity shift; (C) is unsupported by a curve shift; (D) ignores tissue delivery.

Key point: Compare saturation at the same pO_2, not just the apparent direction of a graph.

Q6

Biology · medium

Reproduction / Human gametogenesis chromosome number

Answer: (B) Four, 23.

Given: A human primary spermatocyte is diploid with 46 chromosomes.

Formula/Principle: Two successive meiotic divisions give four haploid products from one primary cell.

Step 1: Meiosis I separates homologues and meiosis II separates sister chromatids.

Step 2: Four spermatids result, each with n=23.

Why: Chromosome number halves while the number of descendant cells quadruples. Meiosis I produces two haploid secondary spermatocytes, each still carrying duplicated chromosomes. Meiosis II separates sister chromatids to give four spermatids with $n = 23$, while fertilisation would restore diploidy later.

Why not the others: (A) stops after meiosis I; (C) retains diploidy; (D) adds an extra division.

Key point: Track cells and chromosomes separately.

Q7

Biology · medium

Genetics And Evolution / Linkage phase and recombinant gametes

Answer: (C) Ab and aB.

Why: The starting coupling-phase parental combinations are AB and ab. Exchanging chromosome segments between the loci creates the mixed combinations Ab and aB. The top chromatid's AB and bottom chromatid's ab are the original combinations. A crossover between the marked loci swaps distal segments, producing Ab and aB; no crossover outside the interval would change this phase.

Why not the others: (A) are both parental; (B) and (D) mix one parental and one recombinant type.

Key point: Establish parental phase before labelling products of crossing over.

Q8

Biology · medium

Biology In Human Welfare / Microbial product and application

Answer: (D) Streptokinase from Streptococcus.

Why: Streptokinase activates plasminogen, promoting fibrin breakdown in clots; it is obtained from streptococcal cultures. The requested function is thrombolysis, not immunosuppression or lipid reduction. Streptokinase does not directly become fibrin; it activates the host plasminogen-to-plasmin system, and plasmin digests the clot. That functional chain distinguishes it from antimicrobial and cholesterol-lowering microbial products.

Why not the others: (A) is immunosuppressive; (B) inhibits cholesterol synthesis; (C) is antibacterial.

Key point: Match mechanism and microbial source to the clinical use.

Q9

Biology · easy

Biotechnology / Bt crop tissue specificity

Answer: (A) Activation and binding of toxin in the larval gut.

Why: Bt protoxin becomes active in the insect gut and binds susceptible gut-cell receptors, leading to membrane damage. Expression by the plant alone does not guarantee that every insect species has the required gut conditions and receptors. In a susceptible alkaline larval gut, the protoxin is activated and can bind specific epithelial receptors. Species lacking compatible receptors need not be harmed merely because the plant expresses the same gene.

Why not the others: (B) concerns plant taxonomy, not larval sensitivity; (C) and (D) assign photosynthetic traits to insects.

Key point: Target specificity depends on the insect's response to the toxin.

Q10

Biology · medium

Ecology / Conservation evidence and endemism

Answer: (B) Loss of X would remove unique lineages from their native habitat.

Why: Endemic species have restricted natural distributions, so destruction of their habitat can cause global loss rather than a local population decline alone. In-situ protection also preserves ecological interactions. Equal richness counts species but does not capture their geographic restriction. If an endemic population disappears from its only habitat, rescue by an unaffected population elsewhere may be impossible.

Why not the others: (A) contradicts the given contrast; (C) confuses count with distribution and threat; (D) overstates ex-situ coverage.

Key point: Richness and irreplaceability are different conservation criteria.

Q11

Biology · easy

Genetics And Evolution / X-linked recessive transmission

Answer: (C) All daughters inherit the father's variant X and are carriers.

Why: A father passes his only X chromosome to every daughter, while the homozygous unaffected mother supplies a normal X. For a fully recessive trait each daughter is heterozygous. All daughters receive the affected father's X rather than his Y. The unaffected homozygous mother's X supplies a functional allele, masking a recessive phenotype while leaving the daughter heterozygous.

Why not the others: (A) and (B) require affected or carrier maternal X; (D) confuses daughters with sons, who receive paternal Y.

Key point: Track the sex chromosome contributed by each parent.

Q12

Biology · medium

Human Physiology / Renal clearance versus reabsorption

Answer: (D) 20 mL/min.

Given: U=40 mg/mL, urine flow V=1 mL/min, plasma P=2 mg/mL.

Formula/Principle: Clearance C=UV/P, a plasma-volume equivalent per unit time.

Step 1: UV=40 mg/min.

Step 2: Divide by 2 mg/mL to obtain 20 mL/min.

Why: This is an excretion-derived clearance; without an independent filtration rate it does not quantify net reabsorption. Clearance has units of volume per time: $(40\,\mathrm{mg/mL})(1\,\mathrm{mL/min})/(2\,\mathrm{mg/mL}) = 20\,\mathrm{mL/min}$. It describes an equivalent plasma volume cleared, not the percentage reabsorbed without a known filtration rate.

Why not the others: (A) and (B) omit a factor; (C) multiplies rather than divides by plasma concentration.

Key point: Preserve units through the quotient.

Q13

Biology · medium

Human Physiology / Neural synapse and receptor response

Answer: (A) Depolarisation of the postsynaptic membrane.

Why: Positive Na^+ entry makes the cell interior less negative, creating an excitatory postsynaptic potential. Reaching an action-potential threshold still depends on size and summation of responses. Opening cation channels makes the inside potential move toward the positive sodium equilibrium potential. An excitatory potential may still be subthreshold, so depolarisation does not guarantee an action potential.

Why not the others: (B) and (D) invoke different ion fluxes; (C) ignores the charge movement across the membrane.

Key point: Predict voltage direction from the sign and direction of ion movement.

Q14

Biology · hard

Cell: Structure And Functions / Competitive versus noncompetitive inhibition

Answer: (B) I competitive; II noncompetitive.

Why: Excess substrate can overcome competition at an active site, retaining Vmax but increasing apparent Km. A pure noncompetitive inhibitor reduces active catalytic capacity, lowering Vmax despite abundant substrate. The control and I curves approach the same upper plateau while I needs more substrate for a given rate, consistent with reversible competition. Curve II's lower plateau cannot be rescued by high substrate under the stated simple model.

Why not the others: (A) cannot explain II's lower plateau; (C) reverses the characteristic patterns; (D) ignores both altered curves.

Key point: Compare saturation plateaus as well as midpoint concentrations.

Q15

Biology · medium

Reproduction / Menstrual hormone sequence

Answer: (C) Degeneration of corpus luteum and fall in progesterone.

Why: Without embryonic maintenance signals the corpus luteum regresses, withdrawing hormonal support from the endometrium. Spiral vessels and tissue then break down and menstruation follows. Absent implantation, there is no sustained signal maintaining the corpus luteum. Falling progesterone removes support for the secretory endometrium, leading to vascular breakdown and shedding in sequence.

Why not the others: (A) maintains the lining; (B) is not a prerequisite and fertilisation normally occurs in the oviduct; (D) contradicts the stated no-implantation cycle.

Key point: Link luteal regression to progesterone withdrawal and lining breakdown.

Chemistry

Method, reasoning and common errors

Q16

Chemistry · hard

Chemical Bonding and Molecular Structure / VSEPR and lone-pair count

Answer: (A) Square planar, two lone pairs.

Why: Six electron domains form an octahedral arrangement. Four bonds leave two lone pairs, and placing them opposite each other minimises lone-pair repulsion; the remaining four bonds lie in one square plane. An octahedral electron-domain arrangement has six positions. Placing the two lone pairs trans to each other maximises their separation and leaves the four fluorines in a square, not a tetrahedron.

Why not the others: (B) confuses molecular with electron geometry; (C) has five domains; (D) also requires five domains.

Key point: Count all electron domains before removing lone-pair positions from the molecular shape.

Q17

Chemistry · medium

Coordination compounds / Chelate stability and ligand replacement

Answer: (B) The chelate effect often gives a favourable entropy change from ligand exchange.

Why: Multidentate ligands form rings with several attachments to the same metal; exchange can liberate more free particles into solution and resist complete dissociation. This comparison assumes similar donor chemistry, so it does not prove every chelate always wins. Chelating ligands bind through two donors to the same metal and form rings. Their dissociation requires breaking coordinated contacts, while exchange can increase the number of free particles and thus favour complex formation entropically.

Why not the others: (A) is not required; (C) contradicts coordination; (D) ignores ligand binding.

Key point: Compare exchange equilibria, not just bond counts.

Q18

Chemistry · easy

The d- & f-block elements / Lanthanoid contraction

Answer: (D) Added 4f electrons shield poorly, increasing effective nuclear attraction.

Why: As protons are added, electrons enter 4f subshells that do not fully screen the increasing nuclear charge from outer electrons, drawing the ionic shell inward. The comparison holds oxidation state fixed at +3. As atomic number grows, proton count increases while the newly added 4f electrons provide relatively weak shielding. The same +3 charge across the graph makes the radius decline meaningful rather than a change in oxidation state.

Why not the others: (A) reverses nuclear change; (B) predicts weak contraction; (C) violates the trivalent label.

Key point: Compare like-charge ions while tracking shielding effectiveness.

Q19

Chemistry · medium

Structure of the Atom / Hydrogen transition wavelength ratio

Answer: (C) 27/5.

Given: The Rydberg wavenumbers are R(1/4−1/9)=5R/36 and R(1−1/4)=3R/4.

Formula/Principle: Wavelength is inverse to wavenumber.

Step 1: Ratio λ32/λ21=(3/4)/(5/36).

Step 2: Simplify to 27/5.

Why: The lower-energy Balmer photon has the longer wavelength. Hydrogen's wavenumber is the difference of inverse-square level terms, not the difference of the level labels. Since $\tilde{\nu}_{32} < \tilde{\nu}_{21}$, the 3→2 wavelength must be longer, checking the ratio's direction.

Why not the others: (A) is the inverse; (B) and (D) confuse level-number ratios with spectral differences.

Key point: Compute energy differences before comparing wavelengths.

Q20

Chemistry · medium

Classification of Elements and Periodicity in Properties / Isoelectronic ionic radii

Answer: (A) $O^{2-} > F^{-} > Na^{+} > Mg^{2+}$.

Why: All have the same electron count and broadly similar electronic arrangement. Nuclear charge increases from O to Mg, pulling that electron cloud closer and shrinking the ion. All four ions have neon-like ten-electron configurations, so electron count is controlled. Increasing nuclear charge pulls essentially the same electron cloud inward, ordering O^{2-} largest through Mg^{2+} smallest.

Why not the others: (B) reverses the trend; (C) swaps O and F; (D) incorrectly places the cations above the anions.

Key point: Hold electron number constant when comparing effective nuclear pull.

Q21

Chemistry · hard

Some Basic Concepts of Chemistry / Aliquot dilution and limiting reagent

Answer: (B) 0.33 g.

Given: Original carbonate=40 mmol; one quarter of diluted solution contains 10 mmol; HCl=15 mmol.

Formula/Principle: Two HCl moles produce one CO_2 mole; molar mass CO_2=44 g/mol.

Step 1: HCl limits reaction to 15/2=7.5 mmol CO_2.

Step 2: Mass=.0075×44=0.33 g.

Why: Dilution leaves total carbonate moles unchanged, but the aliquot captures only a quarter. Dilution changes concentration but conserves the original 40 mmol carbonate; the aliquot contains only 10 mmol. The acid supplies 15 mmol, limiting the reaction to 7.5 mmol CO_2 through the 2:1 stoichiometry.

Why not the others: (A) treats carbonate as limiting; (C) neglects the 2:1 acid ratio; (D) uses all original carbonate.

Key point: Track total, aliquot and stoichiometric limit in order.

Q22

Chemistry · medium

Organic Chemistry – Some Basic Principles and Techniques / Resonance and acidity

Answer: (C) Phenoxide delocalises negative charge into the ring whereas cyclohexoxide lacks that resonance.

Why: Acid strength depends on the stability of the conjugate base. Phenoxide has resonance contributors involving the aromatic π system; the saturated cyclohexyl ring does not provide analogous delocalisation. Deprotonation tests the stability of the *products* of acid dissociation. Phenoxide can distribute negative charge into an aromatic π system, whereas cyclohexoxide remains localised on oxygen in a saturated ring.

Why not the others: (A) assigns aromatic resonance to a saturated ring; (B) denies the principal oxygen-centred structure; (D) describes the wrong leaving species.

Key point: Compare conjugate-base stabilisation rather than only neutral structures.

Q23

Chemistry · easy

Aldehydes, ketones, and carboxylic acids / Esterification equilibrium

Answer: (D) Continuously remove water.

Why: Water is a product; removing it lowers the reaction quotient and the system responds by forming more ester and water. This changes equilibrium composition without asserting a change in equilibrium constant at fixed temperature. With K fixed at unchanged temperature, removing water lowers the product-side reaction quotient. The system then proceeds forward until a new equilibrium composition restores Q=K; the equilibrium constant itself is unchanged.

Why not the others: (A) and (C) add products; (B) removes a reactant, each favouring the reverse direction.

Key point: Distinguish a composition shift from changing K.

Q24

Chemistry · medium

Alcohols, phenols, and ethers / Aryl-alkyl ether cleavage

Answer: (A) Phenol and CH_3I.

Why: After ether oxygen is protonated, iodide attacks the accessible methyl carbon by SN2, breaking the O–CH_3 bond. The aryl C–O bond has partial double-bond character and is not cleaved by ordinary backside attack at sp^2 carbon. Iodide can attack the unhindered methyl carbon after protonation of ether oxygen. Direct SN2 attack at the sp^2 aromatic carbon cannot follow the same trajectory, so the phenolic C–O link remains.

Why not the others: (B) assumes aryl substitution; (C) and (D) change the aromatic skeleton.

Key point: In aryl–alkyl ethers the alkyl side is normally cleaved.

Q25

Chemistry · medium

Organic compounds containing nitrogen / Aniline bromination and protection

Answer: (B) Acetylate –NH_2, brominate, then hydrolyse the amide.

Why: Acyl protection reduces the amino group's strong activation while retaining ortho/para direction; steric effects favour para substitution. Hydrolysis restores –NH_2 after controlled substitution. The acetyl group decreases lone-pair donation into the ring without eliminating directing influence. Once mainly para bromination occurs, hydrolysis restores the amino group while keeping the installed bromine position.

Why not the others: (A) does not selectively moderate activation; (C) worsens multiple substitution; (D) loses the directing group needed for the target.

Key point: Temporarily reduce activation to control substitution count.

Q26

Chemistry · medium

Biomolecules / Double-stranded DNA base pairing

Answer: (C) 20%.

Given: A=30% across both strands.

Formula/Principle: Chargaff pairing gives A=T and G=C for double-stranded DNA.

Step 1: T=30%, leaving 40% for G+C.

Step 2: G=C=20%.

Why: Complementary pairing applies to the combined double-stranded composition. Equal amounts of A and T occupy 60% of all nucleotides, leaving 40% jointly for G and C. Equality of G and C divides that remainder in half; the inference requires a double-stranded paired sample.

Why not the others: (A) halves twice; (B) assigns all bases the A share; (D) gives all the residual share to G.

Key point: Account for both members of each base pair.

Q27

Chemistry · medium

Solutions / Deviation from ideality and interactions

Answer: (D) −15 kPa; stronger unlike attractions.

Why: The ideal total is 0.5×100+0.5×50=75 kPa. Measured minus ideal is 60−75=−15 kPa; a lower escape tendency is consistent with stronger A–B attractions than like-particle interactions. The Raoult benchmark is $0.5(100)+0.5(50)=75\,\mathrm{kPa}$; measured 60 kPa is 15 kPa lower. Stronger unlike attractions are a plausible molecular cause, not a proof of a unique bonding structure.

Why not the others: (A) and (C) reverse the measured sign; (B) assigns the opposite interaction trend, which usually raises vapour pressure.

Key point: Calculate the ideal benchmark before diagnosing an interaction mechanism.

Q28

Chemistry · hard

Electrochemistry / Electrolytic discharge with competing ions

Answer: (A) Cu(s) at cathode and Cl_2(g) at anode.

Why: The supplied preference bars choose $Cu^{2+}+2e^-\rightarrow Cu$ at the negative cathode and $2Cl^-\rightarrow Cl_2+2e^-$ at the positive anode. They explicitly resolve concentrations and overpotential ambiguities that could otherwise alter anode competition. Read the taller cathodic bar for Cu^{2+} reduction and the taller anodic bar for Cl^- oxidation. These measured preferences are stipulated for the cell, avoiding an unjustified universal discharge order that ignores concentrations and overpotential.

Why not the others: (B) incorrectly deposits highly reactive Na; (C) puts reduction at the anode; (D) assigns sulfate a sulfur-dioxide oxidation.

Key point: Read preferences for each electrode, then assign reduction and oxidation separately.

Q29

Chemistry · medium

Chemical Kinetics / Rate response to activation energy

Answer: (B) 1/2.

Given: Equal pre-exponential factors and activation-energy difference RT ln2.

Formula/Principle: $k=A\exp(-E_a/RT)$.

Step 1: Divide the two rate constants: $kY/kX=\exp[-(RT\ln 2)/(RT)]$.

Step 2: $\exp(-\ln 2)=1/2$.

Why: An extra barrier suppresses rate at the same temperature; the unknown common A cancels. The Arrhenius exponent changes by $-\Delta E_a/(RT)=-\ln 2$. Equal prefactors cancel, leaving $e^{-\ln 2}=1/2$; the sign also makes physical sense because a higher barrier slows the reaction.

Why not the others: (A) reverses the sign; (C) ignores barrier change; (D) doubles the exponent.

Key point: Compare Arrhenius rates as a ratio.

Q30

Chemistry · medium

Equilibrium / Reaction quotient after composition change

Answer: (D) Reverse because Qc=4 M^{-1} exceeds Kc.

Given: Current concentrations A=B=1 M and C=4 M.

Formula/Principle: Qc=[C]/([A][B]); if Q>K, reverse reaction lowers Q.

Step 1: Qc=4/(1×1)=4 M^{-1}.

Step 2: Since 4>2, C is consumed initially.

Why: The instantaneous composition has too much product relative to equilibrium. At the stated instant $Q_c = 4\,\mathrm{M}^{-1}$ while $K_c = 2\,\mathrm{M}^{-1}$. Consuming C and generating A and B lowers the product-to-reactant ratio toward K; equal initial reactant concentrations do not imply equilibrium.

Why not the others: (A) and (B) ignore the mass-action expression; (C) reverses the Q>K implication.

Key point: Evaluate Q with the current composition, then compare with K.

Mathematics

Method, reasoning and common errors

Q31

Mathematics · medium

Coordinate Geometry / Tangency of parameterised circle to axis

Answer: (A) 4.

Given: Completing squares yields (x−2)²+(y+3)²=13−c, centre (2,−3).

Formula/Principle: Tangency to x-axis requires radius equal to centre-to-axis distance 3.

Step 1: r²=9.

Step 2: 13−c=9, hence c=4.

Worked derivation:

$$(x-2)^2+(y+3)^2=13-c,\quad r=|-3|=3,\quad 13-c=r^2=9\Longrightarrow c=4.$$

Why: A real nonzero radius is also confirmed. Completing the square identifies the centre before comparing it with the x-axis. The resulting radius is real and nonzero, so the circle has a single contact point.

Why not the others: (B) gives radius 2; (C) zero radius; (D) a larger circle crossing the axis twice.

Key point: Translate the geometric tangency into a radius condition.

Q32

Mathematics · medium

Differentiation / Monotonicity from derivative sign

Answer: (B) (−∞,−1) and (1,∞).

Given: f′(x)=3x²−3.

Formula/Principle: Positive derivative implies local strict increase on intervals.

Step 1: f′>0 when x²>1.

Step 2: This gives x<−1 or x>1; between them f′<0.

Worked derivation:

$$f'(x)=3x^2-3=3(x-1)(x+1)>0\Longleftrightarrow x<-1\text{ or }x>1.$$

Why: Stationary points separate three sign regions, not a single punctured increasing domain. The derivative is negative on the entire middle interval, not merely at a point. The isolated zeros mark changes in monotonicity, so solve the sign inequality over intervals.

Why not the others: (A) is decreasing; (C) includes it; (D) ignores negative derivative throughout (−1,1).

Key point: Make a sign chart rather than checking only derivative zeros.

Q33

Mathematics · medium

Integration / Recover an upper bound from a definite integral

Answer: (C) √3.

Given: u=1+x² transforms the lower and upper bounds to 1 and 1+a².

Formula/Principle: ∫u⁻²du=−1/u.

Step 1: Integral=1−1/(1+a²). Set it equal to 3/4.

Step 2: 1/(1+a²)=1/4, giving a²=3; a>0 selects √3.

Worked derivation:

$$u = 1 + x^2, \quad \int_0^a \frac{2x\,dx}{(1+x^2)^2} = \int_1^{1+a^2} u^{-2}\,du = 1 - \frac{1}{1+a^2} = \frac{3}{4} \Longrightarrow a = \sqrt{3}.$$

Why: The measured area fixes the bound rather than asking for a routine value at supplied endpoints. The upper bound changes under substitution and remains symbolic. The positive-a condition discards the negative square root after solving the resulting equation.

Why not the others: (A), (B), (D) give 1/2, 4/5 and 2/3 respectively.

Key point: Keep the transformed upper limit symbolic until the area condition is applied.

Q34

Mathematics · medium

Relations & functions / Domain of composition with radical and pole

Answer: (D) [2,3).

Given: g must be defined and g(x)≥1 for the square root.

Formula/Principle: Solve 1/(3−x)≥1 with x≠3 by sign intervals.

Step 1: For x>3 the quotient is negative and cannot reach 1.

Step 2: For x<3, multiply by positive 3−x to obtain 1≥3−x, so x≥2.

Worked derivation:

$$g(x) \geq 1,\ x \neq 3; \quad \frac{1}{3-x} \geq 1, \quad x < 3 \Rightarrow x \geq 2; \qquad \mathrm{dom}(f \circ g) = [2, 3).$$

Why: x=2 gives radical zero and is included; x=3 is a pole and excluded. The inner rational expression must exist and then satisfy the outer square-root threshold. Multiplying an inequality without first fixing the denominator's sign could reverse it.

Why not the others: (A) admits x<2; (B) includes the pole and beyond; (C) reverses the bound.

Key point: Impose inner-function definition before outer-function domain.

Q35

Mathematics · easy

Matrices & determinants / Determinant under row replacement

Answer: (A) 5.

Given: One row is augmented by a multiple of another row.

Formula/Principle: Such a row replacement leaves determinant unchanged.

Step 1: Apply $R_2 \to R_2 + 3R_1$.

Step 2: Therefore det B=det A=5.

Worked derivation:

$$\det(B) = \det(R_1, R_2 + 3R_1, R_3) = \det A + 3\det(R_1, R_1, R_3) = 5 + 0 = 5.$$

Why: Multilinearity adds a term with two equal rows, whose determinant is zero. The extra determinant vanishes because it has two identical rows. This explains the invariant rather than merely recalling a row-operation rule.

Why not the others: (B) and (C) incorrectly scale determinant; (D) assumes dependent rows were created.

Key point: Distinguish row replacement from row scaling.

Q36

Mathematics · medium

Probability / Bayes posterior from two draws without replacement

Answer: (B) 3/7.

Given: Priors are P(A)=2/3 and P(B)=1/3; the two-red event has different without-replacement likelihoods.

Formula/Principle: Bayes posterior is P(B)P(RR|B) divided by the sum of both prior-weighted likelihoods.

Step 1: P(RR|A)=(2/3)(1/2)=1/3; P(RR|B)=(3/4)(2/3)=1/2.

Step 2: B's weighted likelihood is 1/6, A's is 2/9; posterior (1/6)/(1/6+2/9)=3/7.

Worked derivation:

$$P(RR \mid A) = \frac{2}{3}\frac{1}{2} = \frac{1}{3}, \quad P(RR \mid B) = \frac{3}{4}\frac{2}{3} = \frac{1}{2}; \quad P(B \mid RR) = \frac{(1/3)(1/2)}{(2/3)(1/3) + (1/3)(1/2)} = \frac{3}{7}.$$

Why: The second draw depends on the first, so multiplying a single-draw red probability by itself would be wrong. Without replacement changes the second-draw probability in each box. The posterior compares prior-weighted two-draw likelihoods, not the raw red counts.

Why not the others: (A) is only the prior; (C) ignores the unequal priors; (D) reverses posterior B with A.

Key point: Form sequential likelihoods before applying Bayes.

Q37

Mathematics · medium

Permutations and Combinations / Identical letters constrained as block

Answer: (D) 840.

Given: MISSISSIPPI has M¹ I⁴ S⁴ P². Treat I⁴ as one block.

Formula/Principle: Distinct permutations of multiset with eight objects equal 8!/(4!2!).

Step 1: The objects are block, M, four S and two P: total 8.

Step 2: 40320/(24×2)=840.

Worked derivation:

$$\text{objects} = \{I^4, M, S^4, P^2\} \text{ with } I^4 \text{ one block}; \quad N = \frac{8!}{4!\,2!} = 840.$$

Why: I's within their contiguous block are identical, so there is no internal 4! factor. The internal order of identical I's contributes only one arrangement. There are eight external objects after blocking, with repeated S and P requiring separate divisors.

Why not the others: (A) fails to divide by P repetition; (B) divides twice; (C) overcounts internal I swaps.

Key point: Count the block once and correct only remaining identical objects.

Q38

Mathematics · medium

Binomial Theorem / Adjacent binomial coefficients

Answer: (C) 7.

Given: Adjacent coefficients C(n,3) and C(n,4) are equal.

Formula/Principle: C(n,r+1)/C(n,r)=(n−r)/(r+1).

Step 1: Set (n−3)/4=1.

Step 2: n=7; indeed C(7,3)=C(7,4)=35.

Worked derivation:

$$\frac{\binom{n}{4}}{\binom{n}{3}} = \frac{n-3}{4} = 1 \Longrightarrow n = 7, \qquad \binom{7}{3} = \binom{7}{4} = 35.$$

Why: The row's two central adjacent coefficients tie only for odd n. Equating adjacent coefficients is a ratio condition; the equality places them symmetrically at the centre of an odd-index row.

Why not the others: (A), (B) and (D) make the ratio different from 1.

Key point: Use the ratio rather than expanding factorials.

Q39

Mathematics · medium

Sequences & series / Term from polynomial partial sum

Answer: (A) 29.

Given: $a_n=S_n-S_{n-1}$.

Formula/Principle: A partial sum includes all earlier terms, so adjacent sums differ by the new term.

Step 1: $S_5=3\times25+10=85$.

Step 2: $S_4=3\times16+8=56$; $a_5=85-56=29$.

Worked derivation:

$$a_5 = S_5 - S_4 = [3(5)^2 + 2(5)] - [3(4)^2 + 2(4)] = 85 - 56 = 29.$$

Why: Substituting n=5 directly gives a sum, not its fifth summand. A partial sum accumulates earlier terms. Taking the difference of adjacent sums isolates the fifth term without assuming the series is arithmetic.

Why not the others: (B), (C) and (D) follow incorrect index or algebra substitutions.

Key point: Difference consecutive partial sums.

Q40

Mathematics · hard

Limit & continuity / Parameter cancellation in exponential limit

Answer: (D) (2,8).

Given: Denominator begins at cubic order $x^3/6$.

Formula/Principle: Compare leading Taylor terms after cancelling lower powers.

Step 1: Numerator quadratic coefficient is $2-a$; finite L requires $a=2$.

Step 2: Its cubic term is $(2x)^3/6=8x^3/6$, divided by $x^3/6$ gives $L=8$.

Worked derivation:

$$e^{2x} = 1 + 2x + 2x^2 + \frac{4}{3}x^3 + O(x^4), \quad e^x = 1 + x + \frac{1}{2}x^2 + \frac{1}{6}x^3 + O(x^4); \quad a = 2,\ L = \frac{4/3}{1/6} = 8.$$

Why: A nonzero quadratic residual would make the quotient diverge like 1/x. A finite two-sided quotient requires the numerator's x^2 term to vanish because the denominator starts at x^3. The remaining leading coefficients then determine the limit.

Why not the others: (A) and (C) choose an uncancelled quadratic; (B) misses the cubing of 2.

Key point: Match the first nonzero orders before taking the ratio.

Q41

Mathematics · medium

Differential equations / Eliminate geometric-family constants

Answer: (B) yy''+(y')²=0.

Given: Differentiate y²=4a(x+b) once to get 2yy'=4a, a constant.

Formula/Principle: Differentiate again to eliminate the constant a; b already disappears after the first derivative.

Step 1: Differentiate 2yy'=4a to get 2[(y')²+yy'']=0.

Step 2: Divide by 2 to obtain yy''+(y')²=0.

Worked derivation:

$$y^2 = 4a(x+b) \Longrightarrow 2yy' = 4a \Longrightarrow 2(y')^2 + 2yy'' = 0 \Longrightarrow (y')^2 + yy'' = 0.$$

Why: Two arbitrary constants require a second-order relation; the product rule creates the squared slope term. The first derivative eliminates the shift b and the second eliminates the constant slope factor a. The second differentiation needs the product rule on yy′.

Why not the others: (A) uses a wrong product-rule sign; (C) drops curvature; (D) would force constant or zero branches.

Key point: Track how each differentiation removes a parameter.

Q42

Mathematics · easy

Three-dimensional geometry / Angle of line with plane

Answer: (C) 2/3.

Given: Plane normal n=(0,0,1) and line direction d=(1,2,2), |d|=3.

Formula/Principle: Sine of line-plane angle is |d·n|/(|d||n|).

Step 1: |d·n|=2.

Step 2: Divide by 3 to get 2/3.

Worked derivation:

$$\hat{\mathbf{n}} = (0,0,1), \quad |\mathbf{d}| = \sqrt{1+4+4} = 3, \quad \sin\theta = \frac{|\mathbf{d}\cdot\hat{\mathbf{n}}|}{|\mathbf{d}|} = \frac{2}{3}.$$

Why: This angle is complementary to the line-normal angle. The line-plane angle is complementary to the angle with the plane normal. The normal component, not the in-plane component, supplies its sine.

Why not the others: (A) uses x component; (B) uses in-plane projection; (D) assumes line normal to plane.

Key point: Use the component perpendicular to the plane.

Q43

Mathematics · medium

Vectors / Collinearity and internal division in three dimensions

Answer: (D) (3, 1:1).

Given: AC=(4,4,4) and AB=(a−1,2,2).

Formula/Principle: A point B on segment AC has AB=t AC with 0<t<1; its segment ratio is t:(1−t).

Step 1: Comparing y components gives 2=4t, so t=1/2.

Step 2: Comparing x gives a−1=4(1/2)=2, hence a=3; ratio is (1/2):(1/2)=1:1.

Worked derivation:

$$\overrightarrow{AC} = (4,4,4), \quad \overrightarrow{AB} = (a-1,2,2) = t\overrightarrow{AC}; \quad t = \frac{1}{2},\ a-1 = 2, \quad AB : BC = t : (1-t) = 1 : 1.$$

Why: The between condition validates the positive internal parameter instead of an extension. The y and z components independently confirm the same internal parameter. Since 0<t<1, the result satisfies the stipulated between condition.

Why not the others: (A) and (C) violate x collinearity; (B) has the correct x but wrong segment ratio.

Key point: Recover the common parameter and then the division ratio.

Q44

Mathematics · medium

Statistics / Mean absolute deviation from grouped frequencies

Answer: (A) 1.

Given: Total count=4 and weighted sum=0+2×2+4=8.

Formula/Principle: Mean deviation about mean is Σf|x−x̄|/Σf.

Step 1: x̄=8/4=2.

Step 2: Weighted absolute deviations=1×2+2×0+1×2=4; divide by 4 to get 1.

Worked derivation:

$$\bar{x} = \frac{1(0)+2(2)+1(4)}{4} = 2, \qquad \mathrm{MD}_{\bar{x}} = \frac{1|0-2|+2|2-2|+1|4-2|}{4} = 1.$$

Why: Deviations are absolute, so opposite signs do not cancel. Absolute deviations cannot cancel by sign, and the denominator is the total frequency four rather than the three distinct values.

Why not the others: (B) forgets frequency division; (C) divides twice; (D) uses the number of distinct values as denominator.

Key point: Use total observation count, not class count.

Q45

Mathematics · hard

Trigonometric functions / Quadratic reduction and periodic root count

Answer: (C) 6.

Given: Let s=sin x and use cos2x=1−2s².

Formula/Principle: Solve the quadratic condition in s, then count the valid angular branches over two periods.

Step 1: 2s²+s−1=0 gives s=1/2 or s=−1.

Step 2: On [0,4π], sin x=1/2 has four solutions and sin x=−1 has two; none coincide, total six.

Worked derivation:

$$s=\sin x,\quad s=1-2s^2 \Longrightarrow (2s-1)(s+1)=0;\quad \#\{s=1/2\}=4,\ \#\{s=-1\}=2,\quad N=6.$$

Why: Neither endpoint is a solution, while the second root is a tangent-level sine value occurring once each period. Two full periods lie in the interval. The level −1 occurs once per period while +1/2 occurs twice; neither endpoint adds an extra root.

Why not the others: (A) omits the −1 branch; (B) misses one period; (D) counts −1 twice per period.

Key point: Count each allowed sine level with its own multiplicity.

Physics

Method, reasoning and common errors

Q46

Physics · medium

Motion in a Straight Line / Motion in a Plane / Piecewise acceleration and displacement

Answer: (A) 12 m.

Given: First-stage final speed=0+2×2=4 m/s.

Formula/Principle: Displacement in each constant-acceleration stage is $v_i t + at^2/2$.

Step 1: First displacement=0+2×4/2=4 m.

Step 2: Second displacement=4×4−1×16/2=8 m; total=12 m.

Worked derivation:

$$v_1 = 0 + (2)(2) = 4\,\mathrm{m/s}, \quad s_1 = \tfrac{1}{2}(2)(2)^2 = 4\,\mathrm{m}; \quad s_2 = (4)(4) + \tfrac{1}{2}(-1)(4)^2 = 8\,\mathrm{m}; \quad s = 12\,\mathrm{m}.$$

Why: Velocity stays nonnegative and reaches zero only at the end, so there is no reverse displacement. Velocity at the stage boundary becomes the initial velocity for the second stage. The final velocity is zero, so no reverse-motion correction is needed.

Why not the others: (B) counts only stage two; (C) only stage one; (D) omits braking loss.

Key point: Carry final velocity from one stage into the next.

Q47

Physics · hard

Laws of Motion / Elevator apparent weight under acceleration reversal

Answer: (B) 480 N.

Given: True weight mg=600 N; observed upward-acceleration normal force=720 N.

Formula/Principle: N=m(g+a) upward and N=m(g−a) downward.

Step 1: a=720/60−10=2 m/s² upward initially.

Step 2: After reversal N=60(10−2)=480 N.

Worked derivation:

$$N_1 = m(g+a) = 720 \Rightarrow a = \frac{720}{60} - 10 = 2\,\mathrm{m/s^2}; \quad N_2 = m(g-a) = 60(10-2) = 480\,\mathrm{N}.$$

Why: The scale reports contact force, which depends on acceleration, not velocity direction. The scale measures normal force, not mg alone. Reversing acceleration at the same magnitude switches the sign of ma regardless of the elevator's velocity.

Why not the others: (A) assumes zero acceleration; (C) ignores reversal; (D) doubles the inferred acceleration.

Key point: Infer acceleration from the first reading before changing its sign.

Q48

Physics · medium

Work, Energy and Power / Energy with dissipative horizontal stage

Answer: (C) 60 m²/s².

Worked derivation:

$$\tfrac{1}{2}v^2 = gh - \mu_k g d = (10)(5) - (0.20)(10)(10) = 30\,\mathrm{m}^2/\mathrm{s}^2, \qquad v^2 = 60\,\mathrm{m}^2/\mathrm{s}^2.$$

Why: Descent provides mgh=50m joules; friction on the horizontal segment removes μmgd=.2×m×10×10=20m joules. The remaining kinetic energy is 30m=(1/2)mv², so v²=60. The sloping segment is frictionless, whereas the shaded horizontal segment dissipates energy. Compute the available height energy and the rough-track loss separately before solving for v.

Why not the others: (A) ignores friction; (B) removes too little; (D) treats loss as final energy.

Key point: Locate where friction acts and conserve energy with its work included.

Q49

Physics · easy

System of Particles and Rotational Motion / Angular acceleration with torque and inertia change

Answer: (D) It remains unchanged.

Given: τ′=2τ and I′=2I.

Formula/Principle: τ=Iα about a fixed axis.

Step 1: α′=τ′/I′=2τ/(2I).

Step 2: α′=α.

Worked derivation:

$$\alpha = \frac{\tau}{I}, \quad \alpha' = \frac{2\tau}{2I} = \alpha.$$

Why: Torque and rotational inertia scale together; their ratio sets acceleration. Torque and inertia change together. The rotational form of Newton's second law depends on their ratio, so doubling both does not double acceleration.

Why not the others: (A) multiplies both changes; (B) changes only torque; (C) changes only inertia.

Key point: Compare the ratio, not either factor alone.

Q50

Physics · medium

Gravitation / Gravity within uniform sphere

Answer: (A) g(R)/2.

Given: Enclosed mass at radius r scales as r³ for uniform density.

Formula/Principle: g(r)=G M(r)/r², hence g(r)∝r inside the sphere.

Step 1: M(R/2)=M(R)/8.

Step 2: Divide by (R/2)²=R²/4, yielding (1/8)/(1/4)=1/2 of surface g.

Worked derivation:

$$M(r) = M(R)\left(\frac{r}{R}\right)^3, \quad g(r) = \frac{GM(r)}{r^2} = g(R)\frac{r}{R}; \qquad g(R/2) = \frac{1}{2}g(R).$$

Why: Outer spherical shells exert zero net field at an internal point. Inside a uniform sphere only enclosed mass contributes net field. Using the full mass at r=R/2 would incorrectly apply the exterior inverse-square law.

Why not the others: (B) uses only inverse-square distance; (C) and (D) ignore enclosed-mass reduction.

Key point: Use enclosed mass, not total planetary mass, inside.

Q51

Physics · medium

Mechanical Properties of Fluids / Stokes terminal speed scaling

Answer: (B) 4.

Given: Material densities, fluid viscosity and g are shared.

Formula/Principle: Balancing effective weight ∝r³ with Stokes drag ∝rv gives terminal speed ∝r².

Step 1: Radius ratio=2.

Step 2: Speed ratio=2²=4.

Worked derivation:

$$(\rho_s - \rho_f)\frac{4}{3}\pi r^3 g = 6\pi\eta r v_t \Longrightarrow v_t = \frac{2(\rho_s - \rho_f)gr^2}{9\eta}, \quad \frac{v_t(2r)}{v_t(r)} = 4.$$

Why: Buoyancy changes the density difference but not the radius exponent under the stated conditions. This scaling assumes Stokes flow and equal material and fluid properties. Buoyancy changes the density difference but cancels in the ratio.

Why not the others: (A) uses linear scaling; (C) uses volume scaling; (D) squares twice.

Key point: Derive scaling from force balance.

Q52

Physics · medium

Thermodynamics / Adiabatic ideal-gas temperature ratio

Answer: (C) 2^(−2/3).

Given: Vf/Vi=2 and γ−1=2/3.

Formula/Principle: For reversible adiabatic ideal-gas change, TV^(γ−1)=constant.

Step 1: Tf/Ti=(Vi/Vf)^(γ−1).

Step 2: Substitute (1/2)^(2/3)=2^(−2/3).

Worked derivation:

$$T_i V_i^{\gamma-1} = T_f V_f^{\gamma-1}, \quad \frac{T_f}{T_i} = \left(\frac{V_i}{2V_i}\right)^{5/3-1} = 2^{-2/3}.$$

Why: The gas cools on adiabatic expansion because it does work without heat input. The process is reversible adiabatic, which is what licenses this power law. Expansion lowers temperature because internal energy supplies the work.

Why not the others: (A) assumes exponent one; (B) predicts heating; (D) uses γ instead of γ−1.

Key point: Use the temperature-volume exponent γ−1.

Q53

Physics · medium

Electric Charges and Fields / Electrostatic Potential / Flux with enclosed charges

Answer: (D) 2q/ε₀.

Given: Algebraic enclosed charge q−q+2q=2q.

Formula/Principle: Gauss's law ∮E·dA=Q_enclosed/ε₀.

Step 1: Cancel the +q and −q contributions in charge sum.

Step 2: Remaining +2q gives positive outward flux 2q/ε₀.

Worked derivation:

$$Q_{\text{enc}} = q - q + 2q = 2q, \qquad \Phi_E = \oint \mathbf{E} \cdot d\mathbf{A} = \frac{2q}{\varepsilon_0}.$$

Why: External charges could reshape the local field but cannot change net closed-surface flux. Gauss's law uses algebraic enclosed charge. External charges and uneven local field strength may alter the pattern on the surface but not its net flux.

Why not the others: (A) cancels all charges; (B) misses a q; (C) adds magnitudes without signs.

Key point: Use algebraic enclosed charge.

Q54

Physics · easy

Electrostatic Potential and Capacitance / Series capacitors and shared charge

Answer: (A) 2CV/3.

Given: Series capacitances are C and 2C.

Formula/Principle: Equivalent capacitance is C_eq=(C×2C)/(C+2C)=2C/3; both series elements carry equal charge magnitude.

Step 1: Total series charge Q=C_eq V.

Step 2: Q=2CV/3 on either capacitor.

Worked derivation:

$$C_{\text{eq}} = \left(\frac{1}{C} + \frac{1}{2C}\right)^{-1} = \frac{2C}{3}, \qquad Q_{\text{series}} = C_{\text{eq}} V = \frac{2CV}{3}.$$

Why: Unequal capacitance creates unequal voltage drops, not unequal series charge. In a simple series chain each capacitor has the same magnitude of charge, even though the voltage across the two capacitances differs.

Why not the others: (B) halves the shared charge; (C) and (D) apply total V directly to an individual capacitor.

Key point: Calculate series equivalent capacitance before charge.

Q55

Physics · medium

Moving Charges and Magnetism / Magnetism and Matter / Magnetic dipole potential energy

Answer: (B) +pB/2.

Worked derivation:

$$U_i = -pB\cos 0^\circ = -pB, \quad U_f = -pB\cos 60^\circ = -\tfrac{1}{2}pB; \qquad W_{\text{ext}} = \Delta U = \tfrac{1}{2}pB.$$

Why: Magnetic potential energy is U=−pBcosθ. It changes from −pB at 0° to −pB/2 at 60°, an increase of pB/2 supplied by an external agent in a slow rotation. The quasistatic qualifier lets external work equal the energy increase without a change in kinetic energy; magnetic field work has the opposite sign.

Why not the others: (A) uses field work rather than external work; (C) treats the final energy as zero; (D) ignores orientation dependence.

Key point: Quasistatic external work equals the change in potential energy.

Q56

Physics · medium

Electromagnetic Induction / Mutual induction signed emf

Answer: (C) −1 V.

Given: M=0.50 H and dI₁/dt=+2 A/s.

Formula/Principle: Signed mutual emf $\mathcal{E}_2 = -M\, dI_1/dt$ under matching orientation convention.

Step 1: Multiply 0.50×2=1 V.

Step 2: Lenz's minus sign gives −1 V.

Worked derivation:

$$\mathcal{E}_2 = -M\frac{dI_1}{dt} = -(0.50\,\mathrm{H})(2\,\mathrm{A/s}) = -1\,\mathrm{V}.$$

Why: The secondary opposes the increase in linked positive flux; the sign depends on the explicitly chosen convention. The given matching positive flux convention fixes the negative sign. Without an orientation convention only the magnitude could be reported unambiguously.

Why not the others: (A) drops Lenz's sign; (B) misses the factor M; (D) misses both.

Key point: State a flux orientation when interpreting emf sign.

Q57

Physics · medium

Ray Optics and Optical Instruments / Lens-plane mirror collimation condition

Answer: (D) 12.5 cm.

Given: First-pass lens image is 15 cm right of lens by 1/f=1/30+1/v.

Formula/Principle: Plane mirror at d reflects a converging point at x=15 into a return-side point at x=2d−15; reverse lens output is parallel if that point is at its 10 cm focal plane.

Step 1: Set 2d−15=10.

Step 2: d=12.5 cm, indeed before x=15 as specified.

Worked derivation:

$$\frac{1}{10} = \frac{1}{30} + \frac{1}{v_1} \Rightarrow v_1 = 15\,\mathrm{cm}; \quad x_{\mathrm{return}} = 2d - 15; \quad x_{\mathrm{return}} = f = 10 \Rightarrow d = 12.5\,\mathrm{cm}.$$

Why: The mirror folds the convergent bundle; its image point supplies the object location for the second lens pass. The first image is virtual to the mirror because it lies behind it at d<15. Reflecting that convergent bundle produces a return-side point at 2d−15; placing that point at the lens focus yields parallel output.

Why not the others: (A), (B), (C) put the return point at −5,5,15 cm, not 10.

Key point: Track both lens traversals and the reflected intermediate point.

Q58

Physics · medium

Wave Optics / Fringe visibility from intensities

Answer: (D) 4/5.

Worked derivation:

$$\mathcal{V} = \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}} = \frac{9-1}{9+1} = \frac{4}{5}.$$

Why: Visibility is (Imax−Imin)/(Imax+Imin), which here is (9−1)/(9+1)=8/10=4/5. A nonzero minimum indicates imperfect contrast, so visibility is below one. The nonzero minimum prevents unit visibility. The sum in the denominator normalises contrast against total intensity rather than using a raw 8-unit difference.

Why not the others: (A) ignores the intensity offset; (B) requires zero minimum; (C) uses the minimum-to-maximum ratio incorrectly.

Key point: Use both extrema in the normalised contrast.

Q59

Physics · hard

Atoms / Hydrogen series transition from wavelength

Answer: (B) n=4.

Given: Final level is 2 and spectral wavenumber is 3R/16.

Formula/Principle: $1/\lambda = R(1/2^2 - 1/n_i^2)$.

Step 1: $1/4 - 1/n_i^2 = 3/16$, so $1/n_i^2 = 1/16$.

Step 2: Positive principal quantum number $n_i = 4$.

Worked derivation:

$$\tilde{\nu} = R\left(\frac{1}{2^2} - \frac{1}{n_i^2}\right) = \frac{3R}{16} \Rightarrow \frac{1}{n_i^2} = \frac{1}{16} \Rightarrow n_i = 4.$$

Why: Energy difference, not ratio of level numbers, determines photon wavenumber. The initial level must exceed the terminating level two for emission; the positive integer solution four satisfies that physical direction.

Why not the others: (A), (C), (D) give different inverse-square differences.

Key point: Rearrange the Rydberg equation before extracting the level.

Q60

Physics · medium

Nuclei / Activity decay in a mixture

Answer: (C) 1:2.

Given: Initial activity A_0 is equal; elapsed time is 4 h.

Formula/Principle: A(t)=A_0(1/2)^(t/T_half) for each isotope separately.

Step 1: First decays over two half-lives to $A_0/4$.

Step 2: Second over one half-life to $A_0/2$, ratio=(1/4):(1/2)=1:2.

Worked derivation:

$$A_1(4\,\mathrm{h}) = A_0\,2^{-4/2} = A_0/4, \qquad A_2(4\,\mathrm{h}) = A_0\,2^{-4/4} = A_0/2, \quad A_1 : A_2 = 1 : 2.$$

Why: Equal starting activity does not imply equal remaining activity with different decay constants. Equal starting activities do not mean equal initial nuclei when half-lives differ. Each activity decays by its own half-life over the same elapsed time.

Why not the others: (A) squares only one ratio; (B) reverses order; (D) ignores elapsed time.

Key point: Compare fractions after the same clock time.